
Clinical Risk Assessment for Type-2 Diabetes

Project Id: AIML-7

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1. Problem Description

Type-2 Diabetes is one of the most common chronic diseases worldwide, and early prediction is critical in preventing severe complications such as heart disease, kidney failure, and nerve damage. The primary objective of this project is to develop a machine learning model that predicts the clinical risk of Type-2 Diabetes based on key health parameters: Age, Body Mass Index (BMI), and Insulin Levels.

Data Acquisition & Cleaning: The project utilizes a clinical dataset (pbl.csv) to simulate real patient scenarios. To ensure data integrity, the system cleans column headers and identifies physiologically impossible "zero" values in BMI and Insulin. These are handled via Median Imputation, ensuring the model is not skewed by missing data.

Preprocessing & Normalization: Categorical risk outcomes are converted into numerical form using Label Encoding. Furthermore, StandardScaler is applied to normalize the data. This is essential because parameters like Insulin often have much higher numerical ranges than Age; scaling ensures all features contribute equally to the final prediction.

Comparative Modeling: While the core logic utilizes a Logistic Regression classifier with "balanced" class weights, the project further extends to a comparative analysis of Decision Trees, Random Forests, and SVM. This allows for the selection of the most robust model for health diagnostics.

Evaluation Strategy: Performance is measured using a Confusion Matrix and ROC Curves.

We place a strategic focus on the Recall Score, as correctly identifying at-risk patients (avoiding False Negatives) is more vital in medical applications than minimizing false alarms. Ultimately, this project demonstrates how machine learning serves as a reliable decision support system for healthcare professionals.

2. Architecture

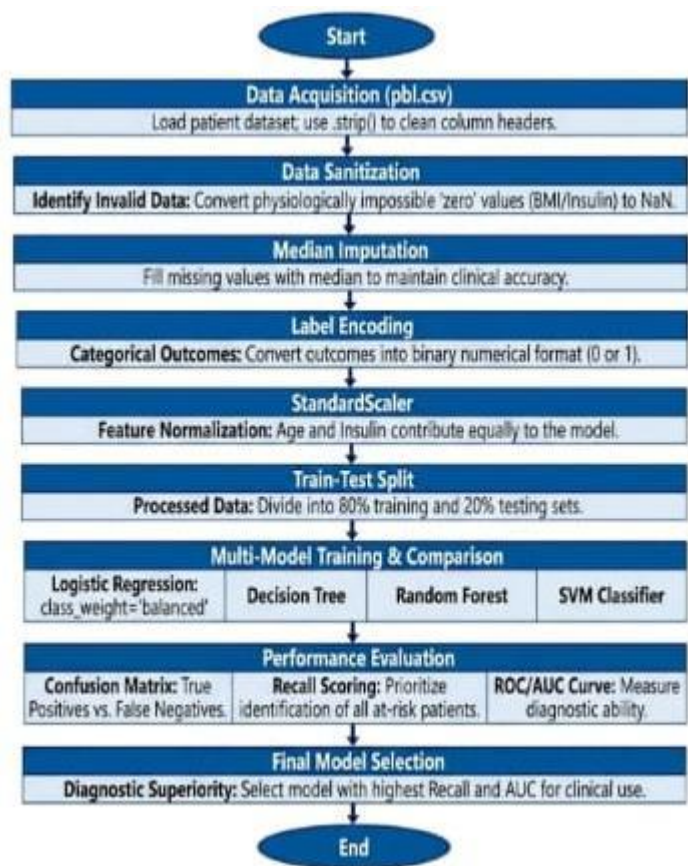


Fig.1 Sample flow diagram.

3. Models Used(Applied/ Planned) Give a brief introduction of the models used

Logistic Regression (LR): Logistic Regression is a linear classifier trained with `class_weight='balanced'` and `max_iter=1000`. It acts as the baseline model. It achieved a validation accuracy of 73.31%.

Decision Tree (DT): A Decision Tree classifier with `max_depth=5` is used to capture non-linear boundaries in the data. It achieved a validation accuracy of 81.06%.

Random Forest (RF): Random Forest with `max_depth=5` is an ensemble of multiple decision trees that reduces overfitting through aggregation. It achieved a validation accuracy of 82.25%.

Gradient Boosting (GB): Gradient Boosting with 100 estimators and learning rate 0.1 builds trees sequentially, each correcting the errors of the previous one. It achieved the highest base validation accuracy of 87.88%.

Support Vector Machine (SVM): SVM with an RBF kernel finds an optimal margin-based hyperplane to separate diabetic and non-diabetic classes. It achieved a validation accuracy of 83.56%.

Ensemble Voting Classifier: All five models are combined using a soft-voting VotingClassifier. It averages the predicted probabilities from all five models and selects the class with the highest combined score, achieving a validation accuracy of 84.31%.

4. Evaluation metrics used

1.Accuracy Score:

Measures the overall correctness of the model by calculating the percentage of correctly predicted cases out of total predictions.

2.Recall Score (Sensitivity):

Measures how many actual diabetic patients are correctly identified by the model. This is the most important metric in medical diagnosis because missing a diabetic patient (False Negative) can be dangerous.

3.Confusion Matrix:

A table used to evaluate classification performance by showing:

True Positives (TP)

True Negatives (TN)

False Positives (FP)

False Negatives (FN)

4.ROC Curve (Receiver Operating Characteristic Curve):

Shows the relationship between True Positive Rate and False Positive Rate at different threshold values.

5.AUC Score (Area Under Curve):

Measures the overall ability of the model to distinguish between diabetic and non-diabetic patients. Higher AUC indicates better performance.

6.Heatmap Visualization:

A graphical representation of the confusion matrix using colors for better understanding of prediction performance.

7.Validation Accuracy:

Used to compare different machine learning models such as Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and SVM during training and testing phases.

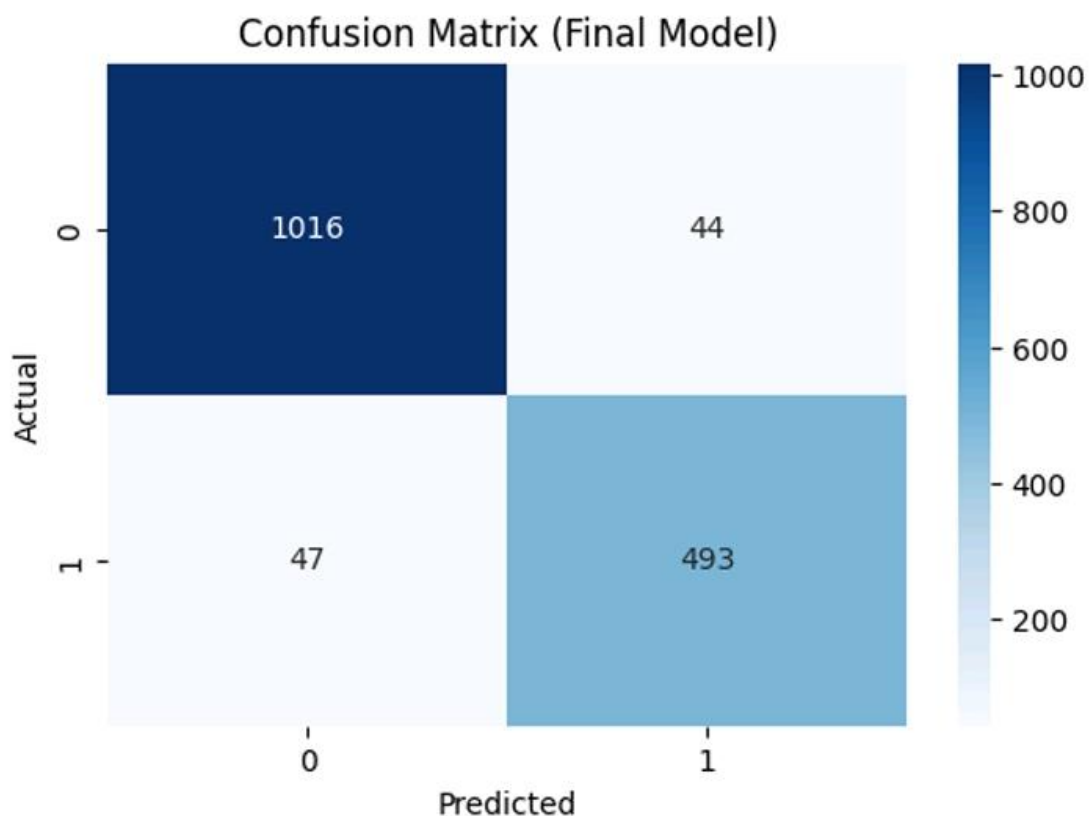
5. Preliminary Results and Visualizations

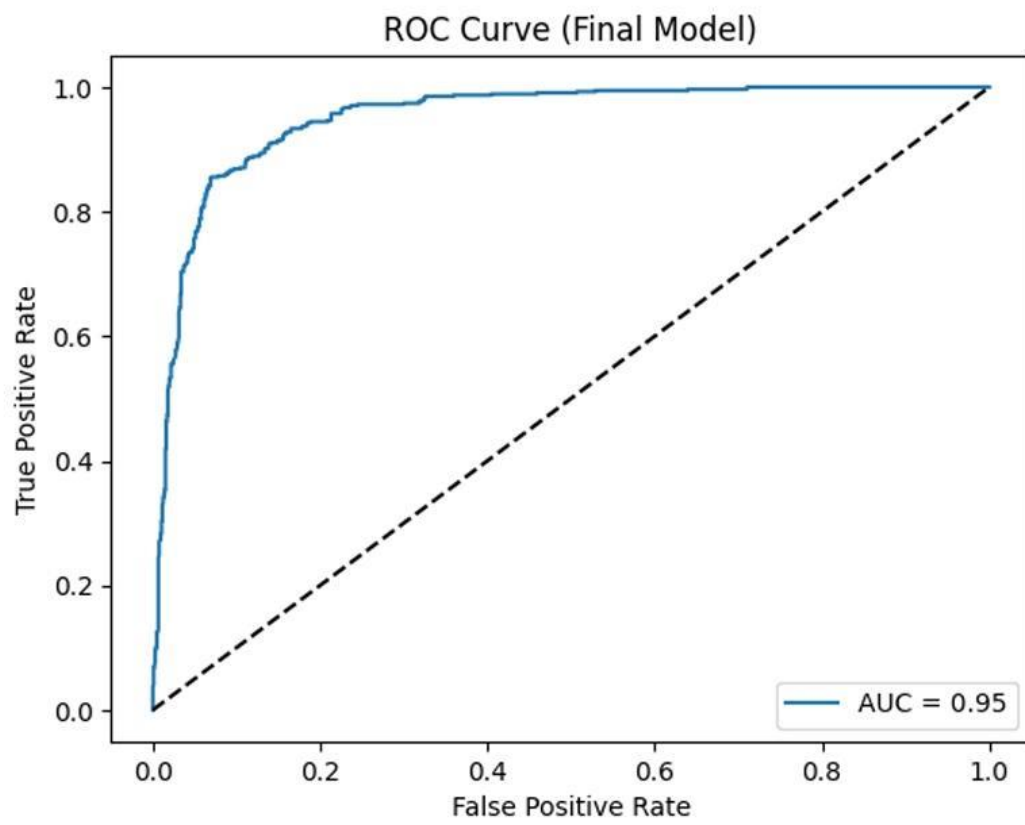
The project achieved high prediction performance using multiple machine learning models for Type-2 Diabetes risk assessment. Among all models, the Tuned Random Forest classifier produced the best results with an accuracy of 94.31%, recall of 91.30%, and F1-score of 91.55%. The ROC curve showed an AUC score of 0.95, indicating excellent discriminative ability between diabetic and non-diabetic patients. The final trained model was saved as `diabetes_model.pkl` for future real-time prediction purposes.

Hyperparameter tuning significantly improved the performance of all models, especially Random Forest, which showed approximately 12% improvement in accuracy after tuning. Gradient Boosting performed best among the baseline models before tuning but was later outperformed by the tuned Random Forest model.

The confusion matrix visualization showed 1016 True Negatives and 493 True Positives with comparatively fewer False Positives and False Negatives, demonstrating strong classification capability. Heatmaps were used for better interpretation of confusion matrix results, while ROC curves were plotted to evaluate model reliability and classification performance.

Feature distribution plots such as histograms were also used to visualize Age, BMI, Glucose, and Insulin distributions within the dataset.





6. References:

- Platform Required:
Visual studio
- Books and Link Sources:
Online Link Sources:
 1. Scikit-learnDocumentation
<https://scikit-learn.org>
 2. PandasDocumentation
<https://pandas.pydata.org>
 3. NumPyDocumentation
<https://numpy.org>
 4. MatplotlibDocumentation
<https://matplotlib.org>